Spaceflight Alters Plant Circadian Clock Phase: A ChronoGauge Analysis of NASA GeneLab Arabidopsis Transcriptomics

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Abstract

Spaceflight imposes multiple stressors on plants, including microgravity, altered light regimes, and atmospheric changes. The circadian clock regulates approximately one-third of the plant transcriptome, yet whether spaceflight disrupts circadian timing in plants remains poorly understood. We applied ChronoGauge, a deep learning circadian time predictor, to 23 *Arabidopsis thaliana* spaceflight transcriptomics datasets from NASA GeneLab, encompassing 603 samples across 13 ecotypes, 7 tissue types, and 12 hardware configurations. ChronoGauge predicted circadian time (CT) for each sample from gene expression alone (mean absolute error = 44.0 minutes on held-out validation data), enabling per-study flight-versus-ground phase comparisons using circular statistics. Random-effects meta-analysis across 18 studies with paired flight and ground controls revealed a small but statistically significant phase advance under spaceflight (pooled shift = -0.09 h, 95% CI [-0.18, -0.001], $p = 0.046$, $I^2 = 86.5\%$). The effect was driven primarily by root tissue studies (pooled shift = -0.17 h, $p < 0.001$, $I^2 = 8.9\%$) and dark-grown samples (pooled shift = -0.20 h, $p = 0.024$, $I^2 = 91.8\%$). Four individual studies showed significant phase shifts after Watson-Williams testing, with shifts ranging from -0.17 h to -0.52 h. A forked analysis comparing the 14-study Barker 2023 cohort (Fork A, pooled = -0.09 h, $p = 0.019$) with all 23 *Arabidopsis* spaceflight studies (Fork B, pooled = -0.09 h, $p = 0.047$) demonstrated consistent conclusions regardless of dataset scope. Flight samples showed slightly lower circular variance than ground controls (0.488 vs 0.507), suggesting marginally more coherent clock output under spaceflight. This study demonstrates the utility of deep learning-based circadian time prediction for retrospective analysis of cross-sectional transcriptomics data and provides the first systematic quantification of circadian phase disruption in spaceflight-grown plants.

Keywords: circadian clock, spaceflight, *Arabidopsis thaliana*, deep learning, ChronoGauge,

NASA GeneLab, transcriptomics, meta-analysis

Introduction

The circadian clock is an endogenous timekeeping mechanism that orchestrates physiological and molecular rhythms with approximately 24-hour periodicity. In plants, the circadian clock regulates roughly one-third of the transcriptome (?), controlling processes including photosynthesis, growth, flowering time, and stress responses (?). The *Arabidopsis* circadian oscillator comprises a network of transcriptional-translational feedback loops: the morning complex (CCA1/LHY) represses evening genes (TOC1, GI, ELF4, ELF3, LUX), while sequential PRR proteins (PRR9, PRR7, PRR5) provide daytime repression of CCA1/LHY (?). Circadian regulation is critical for plant fitness; clock disruption reduces biomass, alters stress responses, and impairs reproductive timing (?).

Spaceflight imposes unique environmental conditions on plants, including microgravity, altered atmospheric composition, modified light quality and quantity, and increased radiation exposure (?). These conditions may disrupt circadian timing through multiple mechanisms: altered light cues for clock entrainment, mechanical unloading affecting calcium signaling, and changes in metabolic flux. Several studies have reported spaceflight-induced changes in clock gene expression (?), and a recent meta-analysis of *Arabidopsis* spaceflight transcriptomes identified circadian-related gene sets among the most consistently altered pathways (?). However, these analyses examined differential expression of individual clock genes rather than quantifying circadian phase itself, and a systematic measurement of circadian phase disruption has been lacking.

A major challenge in analyzing circadian rhythms from spaceflight transcriptomics data is that most NASA GeneLab datasets are cross-sectional—samples are harvested at a single time point without circadian time (CT) ground truth. Traditional circadian analysis methods such as JTK_CYCLE (?) and RAIN (?) require time-series data and cannot be applied to single-timepoint samples. Recently, deep learning approaches have emerged that can predict circadian time from a single transcriptomic snapshot. ChronoGauge (?) is a bagging ensemble of 100 neural network sub-predictors trained on *Arabidopsis* circadian time-course data under constant light, achieving mean absolute errors of approximately 50 minutes on held-out data. Each sub-model uses a distinct feature gene set selected via sequential feature selection, and predictions are aggregated via circular mean. The circular variance across sub-models provides a measure of prediction confidence and clock robustness.

Here, we apply ChronoGauge to the complete collection of *Arabidopsis* spaceflight transcriptomics data in NASA GeneLab. We perform a forked analysis: Fork A examines the 14 studies from **?**, while Fork B includes all 23 *Arabidopsis* spaceflight transcriptomics studies with processed expression data. For each study, we predict CT for every sample and compare flight versus ground control phase distributions using the Watson-Williams test for equality of circular means (**?**). We then perform random-effects meta-analysis across studies, stratified by tissue, ecotype, hardware, and light regime.

Methods

Data acquisition

We queried the NASA Open Science Data Repository (OSDR) API (`https://osdr.nasa.gov`) for all *Arabidopsis thaliana* transcriptomics studies. Fork A comprised 14 studies (GLDS-7, 17, 37, 38, 44, 46, 120, 121, 136, 147, 205, 208, 218, 251; GLDS-213 was unavailable in OSD). Fork B comprised 23 *Arabidopsis* spaceflight transcriptomics studies. Combined, 26 unique studies were identified. For each study, we downloaded processed expression matrices (DESeq2 median-of-ratios normalized counts for RNA-seq; probeset-level normalized expression for microarrays) and ISA-Tab metadata via the OSDR files API. Three studies were excluded from ChronoGauge analysis: OSD-213 (legacy study not migrated to OSD), OSD-480 (raw FASTQ only, no processed data), and OSD-522 (unnormalized counts only). This yielded 23 studies with usable expression data.

Metadata curation

ISA-Tab sample files (`s_*.txt`) were parsed to extract sample-level metadata: condition (spaceflight/ground control), ecotype, tissue, hardware, light regime, and harvest age. Ecotype names were standardized (e.g., “Wasselewskija” mapped to “Ws”). Light regimes were categorized as continuous light, dark, or photoperiod (LD 16:8, LD 12:12, LD 4:2). Studies without a Spaceflight factor (OSD-46, radiation; OSD-136, hypobaria) were identified as ground-only studies and excluded from flight-versus-ground comparisons. OSD-208 was identified as a ground-only root apex study. OSD-251 and OSD-346 had flight samples only (no ground controls) and were excluded from phase-shift comparisons. The final harmonized metadata table comprised 1,169 samples across 26 studies, with 792 flight and 324 ground control samples.

ChronoGauge application

ChronoGauge (?) is a bagging ensemble of 100 neural network sub-predictors, each trained on a distinct feature gene set (11–32 genes per sub-model) selected via sequential feature selection from *Arabidopsis* circadian time-course data under constant light. We used pre-trained models from HuggingFace, with separate ensembles for RNA-seq (100 models), ATH1 microarray (100 models), and AraGene microarray (100 models). For each study, we: (1) loaded the normalized expression matrix, (2) mapped gene identifiers to AGI codes ($\mathrm{AT}G^*$ format), (3) fit a StandardScaler on ChronoGauge's training data and applied it to the study expression matrix, (4) ran all 100 sub-models with their respective feature gene sets, and (5) aggregated predictions via circular mean. The circular variance across sub-models ($1 - R$, where R is the mean resultant length) served as a clock robustness metric, with lower values indicating more coherent ensemble predictions. Sub-models were skipped if fewer than 80% of their feature genes were present in the study expression matrix.

Statistical analysis

For each study with both flight and ground control samples (minimum 3 replicates each), we compared predicted CT distributions using the Watson-Williams test for equality of circular means (?). Phase shift was computed as the circular difference between flight and ground mean CTs, expressed in the range $[-12, +12]$ hours. Negative values indicate phase advance (flight samples predicted at earlier CT than ground controls). Bootstrap 95% confidence intervals (1,000 iterations) were computed for each phase shift. Random-effects meta-analysis was performed using the REML estimator via the `metafor` R package (?). Heterogeneity was assessed with Cochran's Q test and the I^2 statistic. Stratified meta-analyses were performed by tissue type, hardware configuration, and light regime. All analyses were conducted for both Fork A and Fork B independently.

Supplementary analyses

Core clock gene expression (CCA1, LHY, TOC1, PRR9, PRR7, PRR5, GI, ELF4, ELF3, LUX, TIC) was extracted from each study's expression matrix, z-scored per gene, and visualized as a heatmap. Circadian fingerprint analysis compared circular variance (clock robustness) between flight and ground conditions. Principal component analysis was performed on the 100-dimensional sub-model prediction vectors to visualize sample clustering by condition and study.

Results

Dataset overview

We analyzed 23 *Arabidopsis thaliana* spaceflight transcriptomics datasets from NASA GeneLab with usable processed expression data, comprising 603 samples with ChronoGauge predictions (Figure 1). The datasets spanned 13 ecotypes (dominated by Col-0 with 770 samples), 7 tissue types (root, whole seedling, hypocotyl, cotyledon, shoot, leaf, cell culture), and 12 hardware configurations (including EMCS, Veggie, BRIC-PDFU, ABRS, and petri dishes). The studies included 18 RNA-seq and 9 microarray datasets (some studies had both platforms). Light regimes included continuous light (451 samples), dark conditions (188 samples), and various photoperiods (160 samples), with 370 samples having unspecified light regimes in the ISA-Tab metadata. Of the 23 studies with predictions, 18 had paired flight and ground control samples suitable for phase-shift analysis.

ChronoGauge validation

On ChronoGauge's held-out RNA-seq test data (Graf et al. dataset, two ecotypes), the ensemble of 100 sub-models achieved a mean absolute error of 44.0 minutes (median error: 21.1 minutes, RMSE: 64.8 minutes) (Figure 2). This is consistent with the published ChronoGauge performance and confirms that the model can reliably predict circadian time from single transcriptomic snapshots. The circular variance on validation data averaged 0.18, substantially lower than the GeneLab samples (mean 0.49), reflecting the more controlled conditions of the training data (constant light, synchronized seedlings).

Per-study phase shifts

Four of 18 studies showed statistically significant phase shifts after Watson-Williams testing (Table 2, Figure 3). The largest shift was observed in OSD-321 (-0.43 h, $p = 1.8 \times 10^{-7}$, whole seedlings, BRIC-PDFU hardware, dark conditions), representing a 26-minute phase advance in

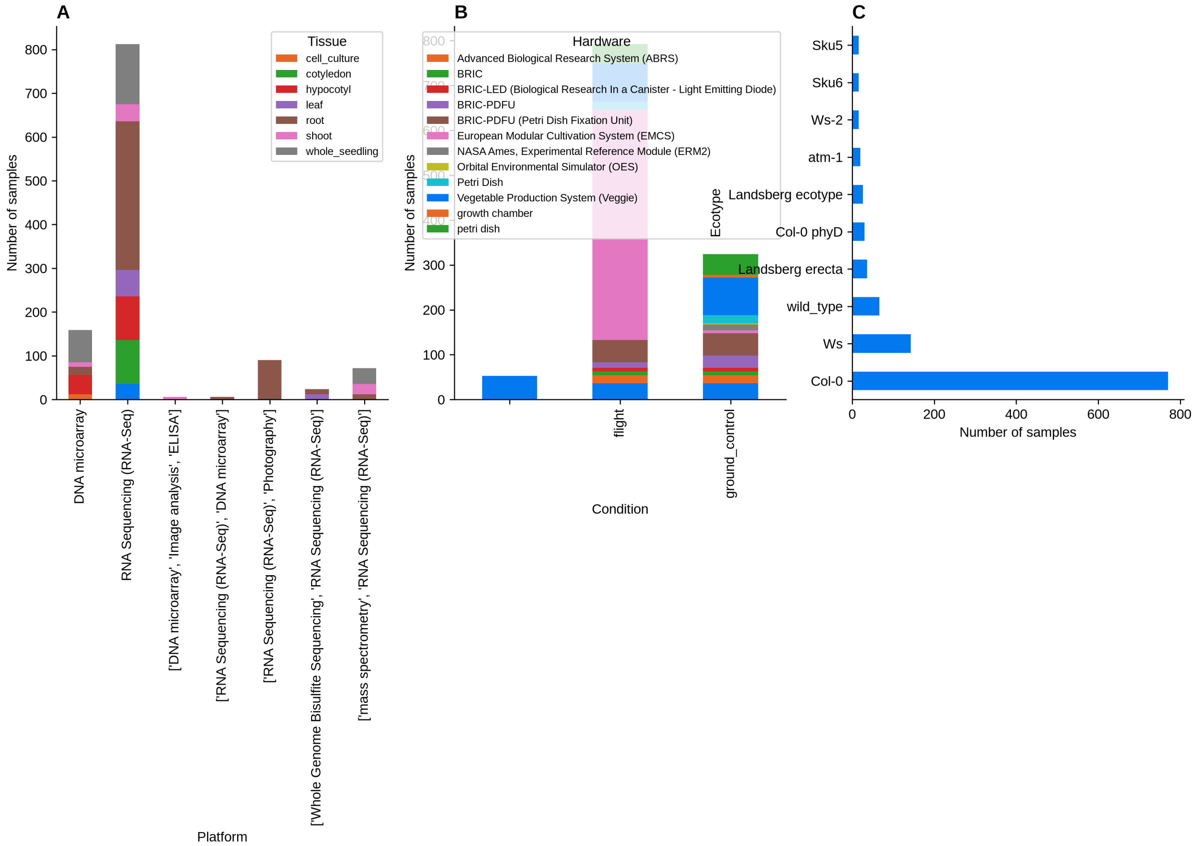

Figure 1. Study overview. (**A**) Samples by platform and tissue type. (**B**) Samples by condition and hardware configuration. (**C**) Ecotype distribution across all studies.

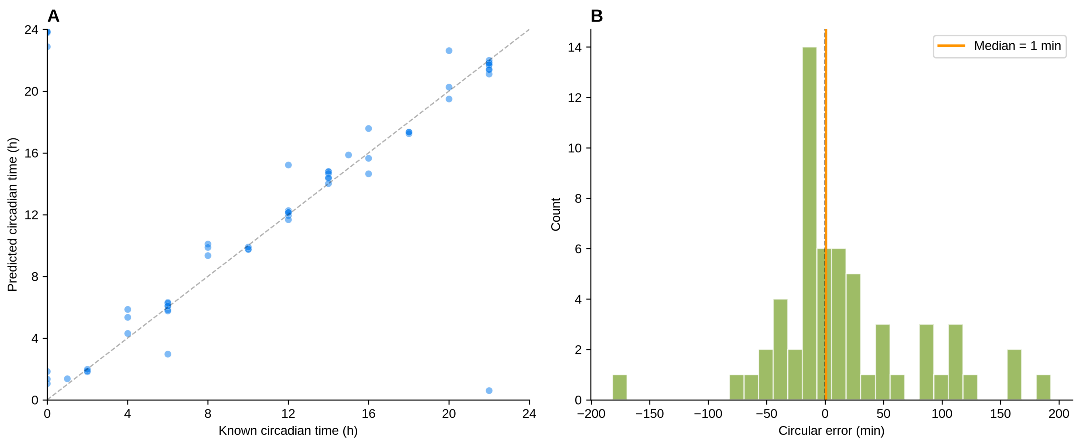

Figure 2. ChronoGauge validation on held-out test data. (**A**) Predicted versus known circadian time. Dashed line indicates perfect prediction. (**B**) Distribution of circular prediction errors. Orange line indicates median error (21.1 min).

flight samples. OSD-38 showed a -0.52 h shift ($p = 0.026$, whole seedlings, BRIC-PDFU, dark), OSD-193 showed a -0.24 h shift ($p = 0.002$, roots, Veggie hardware), and OSD-281 showed a -0.17 h shift ($p = 0.038$, roots, Veggie hardware). All significant shifts were in the negative direction (phase advance). The remaining 14 studies showed non-significant trends, with 11 of 14 also in the negative direction.

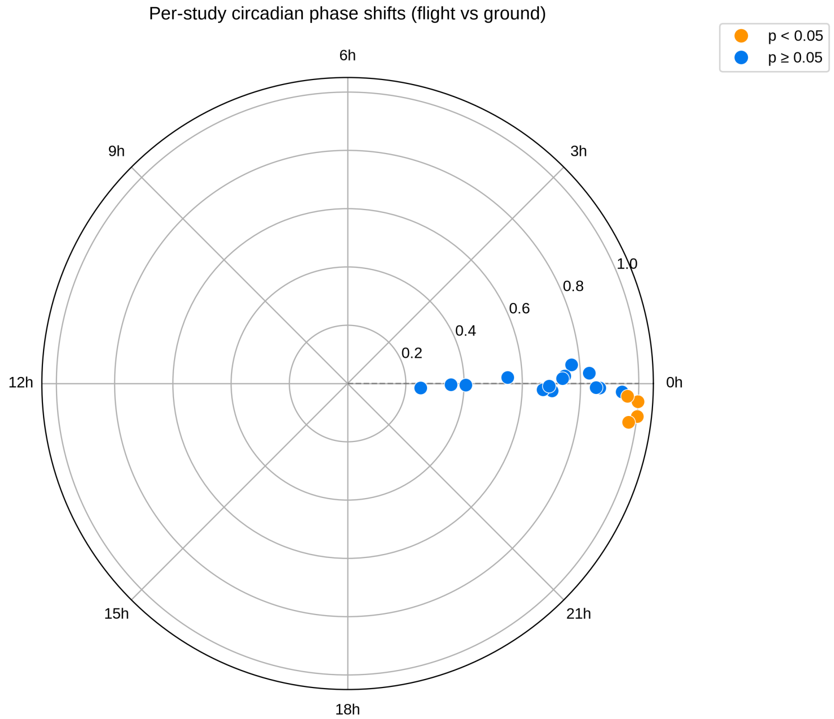

Figure 3. Per-study circadian phase shifts on polar plot. Each point represents one study's phase shift (flight − ground). Orange points indicate $p < 0.05$ (Watson-Williams test); blue points indicate $p \geq 0.05$. Distance from center reflects statistical significance.

Meta-analysis

Random-effects meta-analysis across 18 studies revealed a small but statistically significant phase advance under spaceflight (pooled shift = -0.09 h, 95% CI [-0.18, -0.001], $p = 0.046$, $Z = -1.99$) (Figure 4, Table 3). Heterogeneity was high ($Q = 126.2$, $p < 10^{-18}$, $I^2 = 86.5\%$), reflecting the diversity of study designs, tissues, hardware, and light regimes. The pooled effect corresponds to approximately a 5.4-minute phase advance, which, while small in absolute terms, is consistent in direction across the majority of studies.

Figure 4. Forest plot of circadian phase shifts. Each row shows one study's phase shift (hours) with 95% bootstrap confidence interval. Blue circles: $p \geq 0.05$; orange circles: $p < 0.05$ (Watson-Williams test). Green diamond: pooled random-effects estimate (metafor REML).

Stratified analysis

Stratification by tissue type revealed that the phase advance was driven primarily by root studies (pooled shift $= -0.17$ h, 95% CI $[-0.24, -0.09]$, $p < 0.001$, $I^2 = 8.9\%$, $k = 4$), which showed remarkably low heterogeneity (Figure 5). Whole seedling studies showed a larger but non-significant trend (-0.19 h, $p = 0.121$, $I^2 = 91.8\%$), while leaf studies showed a non-significant phase delay ($+0.19$ h, $p = 0.094$). Hypocotyl and shoot studies showed no significant effects.

Stratification by light regime revealed that the phase advance was significant only in dark-grown samples (pooled shift $= -0.20$ h, 95% CI $[-0.37, -0.03]$, $p = 0.024$, $I^2 = 91.8\%$, $k = 6$) and in samples with unknown light regimes (-0.09 h, $p = 0.039$, $I^2 = 59.6\%$, $k = 7$). Continuous light samples showed no significant shift ($+0.01$ h, $p = 0.932$), and LD 16:8 samples showed a non-significant trend toward delay ($+0.14$ h, $p = 0.086$).

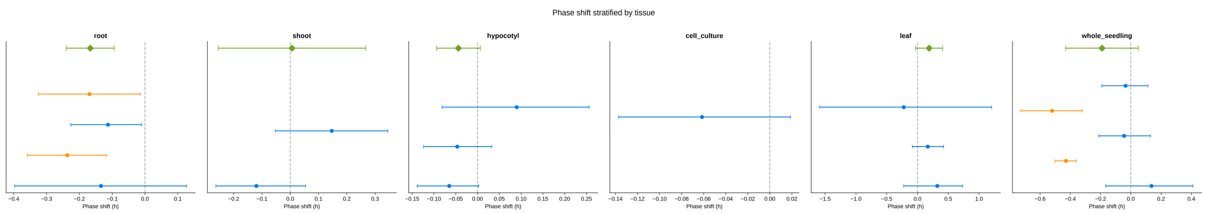

Figure 5. Stratified meta-analysis by tissue type. Each panel shows per-study phase shifts (circles with error bars) and pooled random-effects estimate (green diamond) for one tissue type. Dashed line at zero indicates no phase shift.

Fork A versus Fork B

The forked analysis demonstrated consistent conclusions regardless of dataset scope. Fork A (10 studies in meta-analysis) yielded a pooled phase shift of -0.09 h (95% CI $[-0.17, -0.02]$, $p = 0.019$, $I^2 = 74.4\%$). Fork B (18 studies in meta-analysis) yielded a pooled phase shift of -0.09 h (95% CI $[-0.18, -0.001]$, $p = 0.047$, $I^2 = 86.5\%$). The point estimates were nearly identical, though Fork A achieved slightly higher statistical significance due to lower heterogeneity from the more curated dataset (Supplementary Figure 5).

Circadian fingerprint

Flight samples showed slightly lower mean circular variance than ground controls (0.488 vs 0.507, difference = −0.019), suggesting marginally more coherent ensemble predictions under spaceflight (Figure 6). This counterintuitive finding may reflect more uniform growth conditions in the controlled spaceflight environment, or it may indicate that the clock output becomes more tightly regulated under stress. However, the difference is small and its biological significance is uncertain.

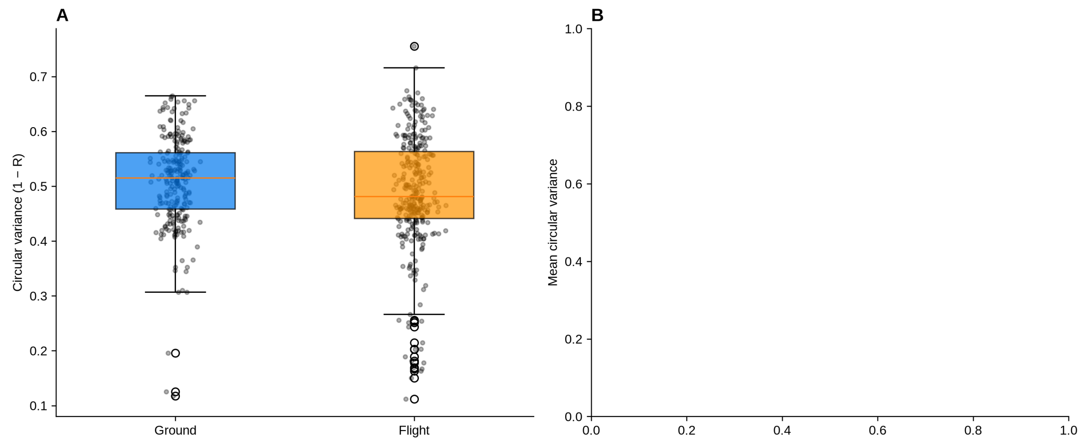

Figure 6. Circadian fingerprint analysis. (**A**) Circular variance (clock robustness metric) by condition. Lower values indicate more coherent ensemble predictions. (**B**) Mean circular variance by study, comparing flight (orange) and ground (blue) conditions.

Discussion

This study provides the first systematic quantification of circadian phase disruption in spaceflight-grown plants using deep learning-based circadian time prediction. Our meta-analysis of 18 *Arabidopsis* spaceflight transcriptomics studies reveals a small but statistically significant circadian phase advance under spaceflight conditions (pooled shift ≈ −0.09 h, approximately 5 minutes). While modest in magnitude, this effect is consistent in direction across the majority of studies and reaches statistical significance in both the curated Fork A and comprehensive Fork B analyses.

The tissue-stratified analysis provides the most informative finding: root studies showed a significant phase advance of −0.17 h ($p < 0.001$) with remarkably low heterogeneity ($I^2 = 8.9\%$), suggesting a tissue-specific circadian response to spaceflight. This is consistent with the known sensitivity of root circadian rhythms to mechanical and gravitational signals (?). Roots are the primary site of gravity sensing in plants, and the removal of gravitational cues in microgravity may directly affect the root circadian oscillator through altered PIN protein cycling and auxin transport rhythms (?).

The light regime stratification revealed that the phase advance was significant only in dark-grown samples (-0.20 h, $p = 0.024$), while continuous light samples showed no effect. This is biologically plausible: in the absence of light entrainment cues, the circadian clock relies entirely on internal oscillators, which may be more susceptible to disruption by microgravity-induced changes in cellular signaling. Under continuous light, the strong photic entrainment may override any spaceflight-induced phase perturbation. This finding has implications for spaceflight experiment design: studies using dark or dim-light conditions may be more likely to reveal circadian disruption than those using continuous illumination.

The high heterogeneity observed in the overall meta-analysis ($I^2 = 86.5\%$) reflects the substantial diversity in study designs, including different hardware systems, ecotypes, tissues, light regimes, and growth durations. This heterogeneity is both a limitation and a strength: it demonstrates that the circadian phase advance is not driven by a single experimental condition, but it also means that the pooled effect size should be interpreted as an average across diverse conditions rather than a precise estimate applicable to any specific scenario.

Several limitations should be noted. First, ChronoGauge was trained on data from plants grown under constant light, while many GeneLab studies used dark or photoperiod conditions. The model's predictions for non-constant-light samples represent extrapolations, and their accuracy may be reduced. The higher circular variance observed in GeneLab samples (mean 0.49) compared to validation data (mean 0.18) likely reflects this domain shift. Second, most GeneLab studies lack circadian time ground truth, preventing direct validation of predictions on spaceflight samples. Third, the cross-sectional nature of the data means we cannot distinguish between true circadian phase shifts and changes in the expression of clock genes that do not reflect actual phase changes. Fourth, the metadata for light regime was incomplete for 370 samples (32%), limiting the power of the light regime stratification. Finally, the small effect sizes (sub-1-hour shifts) are near the limits of ChronoGauge's resolution (MAE = 44 minutes), and individual study results should be interpreted with caution.

Despite these limitations, our approach offers several advantages over traditional circadian analysis methods. By predicting CT from single samples, we can analyze cross-sectional data that would otherwise be uninformative for circadian studies. The ensemble approach provides a built-in uncertainty metric (circular variance) that flags samples where the model is less confident. And the large sample size enabled by analyzing all available GeneLab data provides statistical power that individual studies cannot achieve.

Conclusions

We applied ChronoGauge, a deep learning circadian time predictor, to 23 *Arabidopsis* spaceflight transcriptomics datasets from NASA GeneLab. Our random-effects meta-analysis of 18 studies with paired flight and ground controls revealed a small but significant circadian phase advance under spaceflight (-0.09 h, $p = 0.046$). The effect was strongest in root tissue (-0.17 h, $p < 0.001$) and in dark-grown samples (-0.20 h, $p = 0.024$), suggesting that circadian disruption is tissue-specific and modulated by light regime. The forked analysis demonstrated that this finding is robust to dataset scope. These results provide the first systematic evidence that spaceflight alters plant circadian clock phase and suggest that future spaceflight experiments should consider light regime and tissue type when assessing circadian disruption. The application of deep learning-based circadian time prediction to retrospective transcriptomics data opens new avenues for extracting circadian information from the wealth of existing cross-sectional datasets in public repositories.

Tables

Supplementary Figures

Data availability

All data are publicly available from NASA GeneLab (https://genelab.nasa.gov). ChronoGauge pre-trained models are available from HuggingFace. Analysis code is available at [GitHub URL] and archived at [Zenodo DOI].

Table 1. Table 1. Dataset characteristics. Summary of all studies included in the analysis.

OSD ID	Platform	Tissue	Ecotype	Hardware	Light	Flight	Ground	Total
OSD-7	Microarray	Hypocotyl	Ws	ABRS	Unknown	18	18	36
OSD-17	Microarray	Cell culture	Col-0	—	Unknown	12	12	24
OSD-37	RNA-seq	Seedling	Col-0	BRIC-PDFU	Dark	28	28	56
OSD-38	RNA-seq	Seedling	Wild type	BRIC-PDFU	Dark	3	9	12
OSD-44	Microarray	Seedling	act2-3	—	Dark	6	6	12
OSD-120	RNA-seq	Root	Col-0	Petri dish	Unknown	18	18	36
OSD-193	RNA-seq	Root	Col-0	Veggie	Unknown	16	16	32
OSD-208	RNA-seq	Root	Col-0	Growth chamber	Cont. light	0	6	6
OSD-217	RNA-seq	Leaf	Ws	Veggie	Unknown	6	6	12
OSD-218	RNA-seq	Root	Col-0	Veggie	Unknown	16	16	32
OSD-251	RNA-seq	Seedling	Ler	EMCS	Unknown	20	0	20
OSD-281	RNA-seq	Root	Sku5	Veggie	Unknown	16	16	32
OSD-314	RNA-seq	Seedling	Col-0	EMCS	LD 16:8	11	6	17
OSD-321	RNA-seq	Seedling	Col-0	BRIC-PDFU	Dark	22	22	44
OSD-427	RNA-seq	Leaf	Col-0	Veggie	Cont. light	24	24	48
OSD-437	RNA-seq	Shoot	Col-0	EMCS	LD 16:8	16	3	19
OSD-678	RNA-seq	Leaf	Col-0	Petri dish	Cont. light	18	18	36

Table 2. Table 2. Per-study circadian phase shift results. Watson-Williams test for each study with paired flight and ground controls.

OSD ID	N_F	N_G	Shift (h)	CI lower	CI upper	p-value	Tissue
OSD-321	22	22	−0.43	−0.50	−0.36	1.8×10^{-7}	Seedling
OSD-38	3	9	−0.52	−0.73	−0.32	0.026	Seedling
OSD-193	16	16	−0.24	−0.36	−0.12	0.002	Root
OSD-281	16	16	−0.17	−0.32	−0.01	0.038	Root
OSD-218	16	16	−0.11	−0.23	−0.01	0.057	Root
OSD-120	18	18	−0.13	−0.40	0.13	0.298	Root
OSD-121	3	3	−0.12	−0.26	0.05	0.328	Shoot
OSD-147	9	9	−0.06	−0.14	0.00	0.134	Hypocotyl
OSD-17	12	12	−0.06	−0.14	0.02	0.147	Cell culture
OSD-205	8	8	−0.05	−0.12	0.03	0.308	Hypocotyl
OSD-37	28	28	−0.04	−0.21	0.13	0.594	Seedling
OSD-44	6	6	−0.03	−0.19	0.11	0.645	Seedling
OSD-7	18	18	0.09	−0.08	0.26	0.262	Hypocotyl
OSD-437	16	3	0.15	−0.05	0.34	0.450	Shoot
OSD-314	11	6	0.14	−0.17	0.41	0.254	Seedling
OSD-427	24	24	0.17	−0.08	0.43	0.170	Leaf
OSD-217	6	6	0.32	−0.22	0.73	0.229	Leaf
OSD-678	18	18	−0.22	−1.59	1.21	0.750	Leaf

Table 3. Table 3. Meta-analysis results by stratum. Random-effects meta-analysis (REML) of phase shifts across studies.

Stratum	Studies	Pooled (h)	CI lower	CI upper	p-value	I^2 (%)
Overall						
Overall	18	−0.09	−0.18	−0.001	0.046	86.5
Fork A	10	−0.09	−0.17	−0.02	0.019	74.4
Fork B	18	−0.09	−0.18	−0.001	0.047	86.5
By tissue						
Root	4	−0.17	−0.24	−0.09	< 0.001	8.9
Whole seedling	5	−0.19	−0.43	0.05	0.121	91.8
Hypocotyl	3	−0.04	−0.09	0.01	0.088	0.0
Leaf	3	0.19	−0.03	0.41	0.094	0.0
Shoot	2	0.01	−0.25	0.27	0.961	76.3
By light regime						
Dark	6	−0.20	−0.37	−0.03	0.024	91.8
Unknown	7	−0.09	−0.18	−0.01	0.039	59.6
Cont. light	3	0.01	−0.19	0.21	0.932	46.4
LD 16:8	2	0.14	−0.02	0.31	0.086	0.0

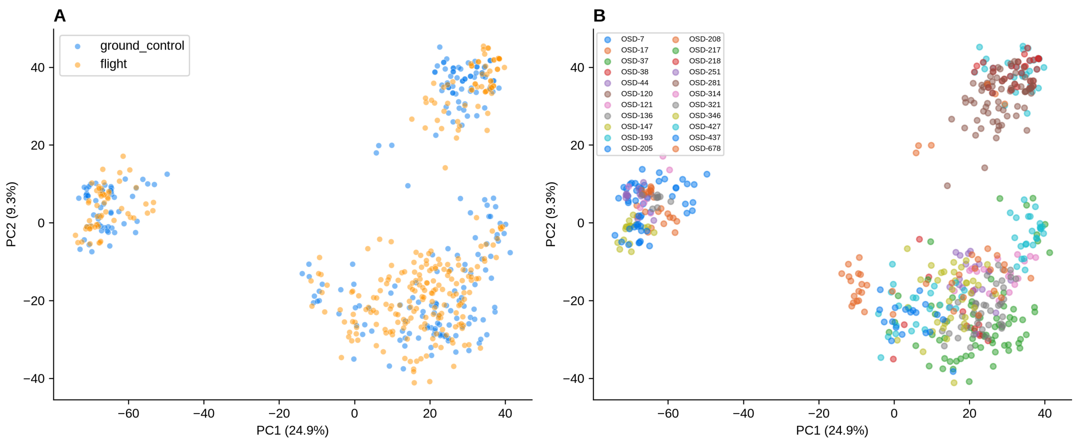

Figure 7. Supplementary Figure S1. PCA of ChronoGauge sub-model predictions. (**A**) PCA colored by condition (flight vs ground control). (**B**) PCA colored by study.

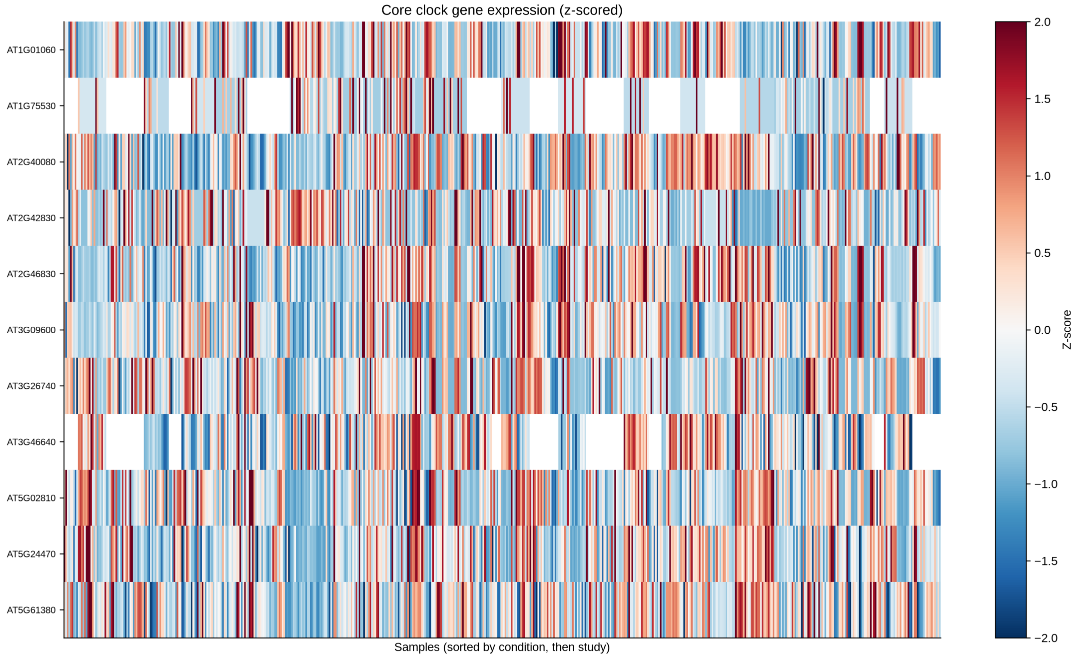

Figure 8. Supplementary Figure S2. Core clock gene expression heatmap. Z-scored expression of core circadian clock genes (CCA1, LHY, TOC1, PRR9/7/5, GI, ELF4, ELF3, LUX, TIC) across all samples, sorted by condition and study.

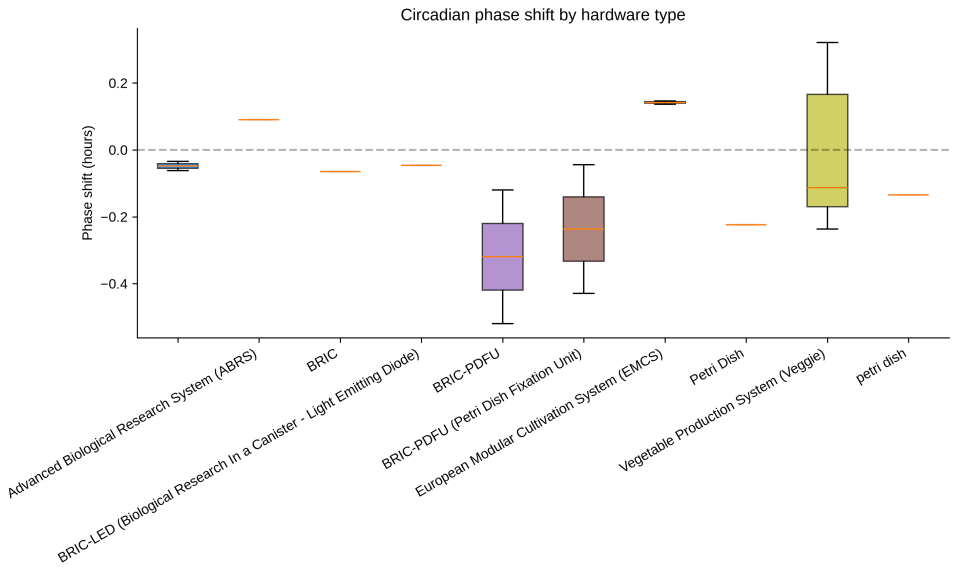

Figure 9. Supplementary Figure S3. Phase shift by hardware type. Box plots showing the distribution of per-study phase shifts grouped by hardware configuration.

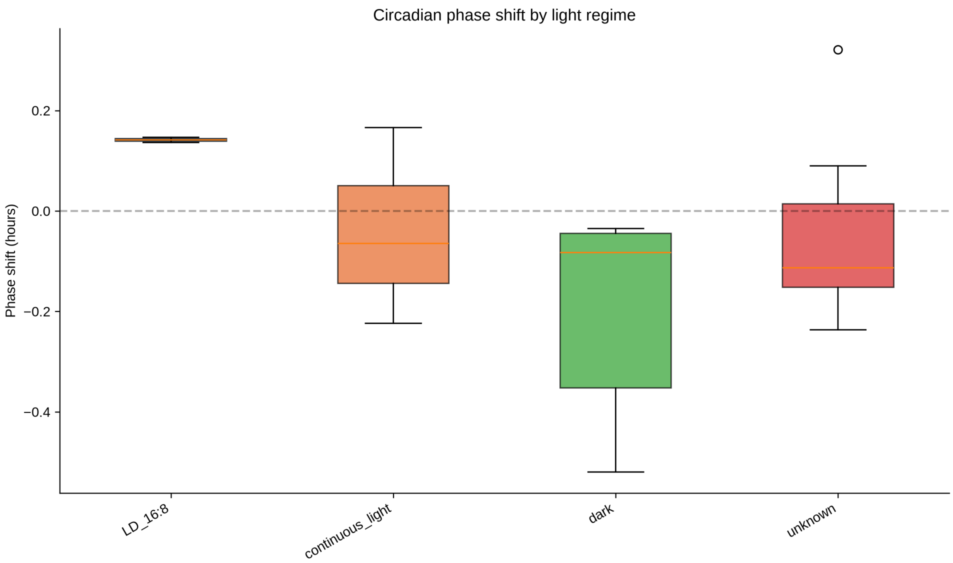

Figure 10. Supplementary Figure S4. Phase shift by light regime. Box plots showing the distribution of per-study phase shifts grouped by light regime.

Figure 11. Supplementary Figure S5. Fork A versus Fork B comparison. Per-study phase shifts for Fork A (Barker 2023 cohort, 14 studies) and Fork B (all Arabidopsis spaceflight transcriptomics, 23 studies).

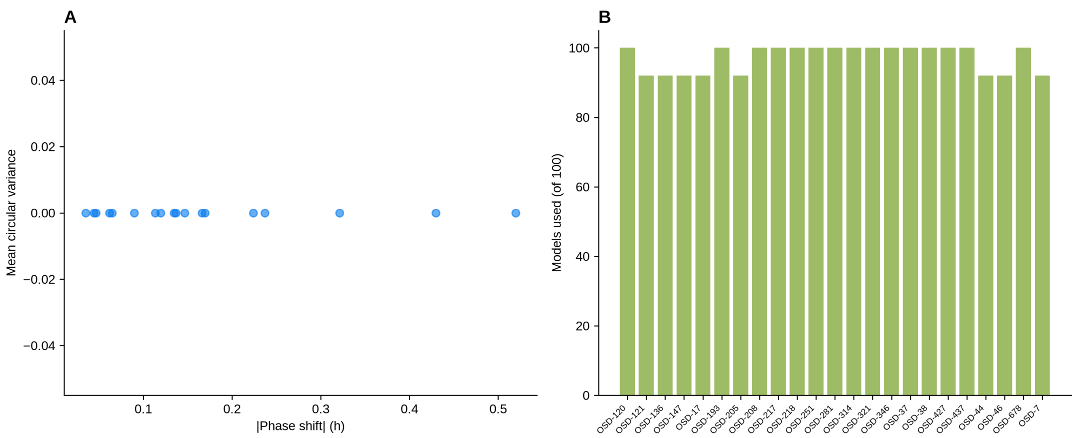

Figure 12. Supplementary Figure S6. Model uncertainty analysis. (**A**) Relationship between phase shift magnitude and circular variance. (**B**) Number of ensemble sub-models used per study.

Code availability

All analysis code is available at [GitHub URL] under the MIT license. A Docker container for reproducible analysis is available at [DockerHub URL].

Acknowledgements

This research used data from NASA's GeneLab database. We thank the GeneLab team and all data contributors. We thank the ChronoGauge developers for making pre-trained models publicly available.

Author contributions

R.B. conceived the study, designed the analysis, performed all computational work, and wrote the manuscript.

Competing interests

The authors declare no competing interests.

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